Table 4: The AUC scores of detectors on text generated by different methods. "–" indicates the case where no evasion method is used. 'AVG' represents the average AUC scores across detectors.

Method	GPT3-D	OpenAI-D	DetectGPT	AVG
–	0.908	0.789	0.834	0.844
Human-ICE	0.918	0.825	0.847	0.863
SICO-Para	0.239	0.488	0.149	0.292
w/o feature	+0.106	+0.072	+0.051	+0.076
w/o ICE	+0.359	+0.133	+0.411	+0.301
w/o OPT	+0.361	+0.154	+0.364	+0.293

For different LLMs, we adopt Vicuna-13B (Chiang et al., 2023), an open-source chatbot fine-tuned from LLaMa (Touvron et al., 2023). Results in Table 5 show that SICO maintains a high degree of detection evasion performance, regardless of proxy detectors or LLMs. In most cases, SICO manages to reduce the AUC of detectors by approximately 0.4. More results of using different LLMs can be found in Appendix B.4.

Table 5: The AUC scores of SICO using different proxy detectors and LLMs on writing task. The first line indicates the performance without applying any evasion method.

LLM	Proxy Detector	GPT3-D	GPT2-D	GPTzero	OpenAI-D	DetectGPT	Log-Rank
-	-	0.908	0.848	0.779	0.789	0.834	0.914
GPT-3.5	GPT3-D	0.239	0.332	0.290	0.488	0.149	0.147
GPT-3.5	GPTzero	0.549	0.143	0.299	0.458	0.429	0.493
GPT-3.5	DetectGPT	0.291	0.361	0.244	0.483	0.174	0.173
Vicuna-13B	GPT3-D	0.393	0.777	0.380	0.690	0.337	0.375

5.3 Examples of SICO text

Table 6: Fake reviews generated by SICO. The first line shows the task input of review generation.

Object: Sushi Station **Keywords**: revolving sushi bar, affordable, krab mix, creamy sauce, fresh menu. **Sentiment**: Positive	$\mathcal{P}_{\text{AI}}$
AI: If you're a sushi lover, you have to check out Sushi Station! This revolving sushi bar is not only affordable, but it offers a fresh and delicious menu that will leave you craving more. My personal favorite is the krab mix sushi with its creamy sauce - it's simply heavenly. The staff is friendly and attentive, and the atmosphere is perfect for a casual night out with friends or family. Don't hesitate to give Sushi Station a try - you won't be disappointed!	0.99
SICO-Para: Sushi enthusiasts, please take note: Sushi Station provides an unbeatable revolving sushi bar experience at wallet-friendly prices. Their menu is of the freshest quality and tastes nothing short of delicious. My top pick has to be the krab mix sushi; its creamy sauce is pure bliss. Additionally, their friendly staff provided excellent service, while the casual ambiance proved ideal for a relaxed night out with friends or family. Don't hesitate to give Sushi Station a try, you won't regret it!	0.00
SICO-Gen: I recently tried out Sushi Station's revolving sushi bar and I must say, it was an amazing experience. The affordable pricing along with the wonderful krab mix and creamy sauce they offer is what really made it stand out from the rest. The menu was constantly updated with fresh and delicious options making it hard to choose just one. If you're looking for a great sushi experience, Sushi Station is definitely worth a visit.	0.04

Table 6 presents the fake reviews generated by SICO-Gen and SICO-Para. The generated text shows high readability and fulfill the task's goal, successfully mentioning all keywords and generating positive reviews of the specified object. The AI probability, denoted as $\mathcal{P}_{\text{AI}}$ in the table, is determined by the GPT3-D. More examples are shown in Appendix H.

6 Defensive Methods for SICO

6.1 Training Detectors with SICO Data

To defend against SICO, a straightforward approach is to train a detector on SICO text. We proposed an ensemble method, which utilizes two detectors trained separately to identify original AI-generated text and SICO text. The final AI probability for an input is calculated using the highest one from the two detectors, based on the assumption that both detectors should assign a low probability to human-written text. We avoided training a single detector with SICO text augmentation because our experiments indicated that this approach reduces the model's ability to identify the original AI-generated text.

To test the effectiveness of this method, we obtain the SICO detector by fine-tuning a RoBERTa model with 5k SICO examples and 5k human-written examples. The human-written examples are sampled from WebText training set (Radford et al., 2019). The SICO examples are generated by using three different SICO-Para prompts to rewrite AI-generated text from the GPT2 output dataset (Solaiman et al., 2019). The SICO prompts used for detector training is entirely different from the one used for attack. We adopted the "GPT2-D" as the original AI-generated text detector. The results in Table 7 show that the ensemble approach significantly improves detection capabilities against SICO text, while maintaining the ability to identify the original AI-generated text. We believe this approach can be easily adapted to defend against other types of attacks, such as DIPPER, by training and incorporating detectors with adversarial attack.

Table 7: AUC scores of ensemble detector against normal AI-generated text and SICO text. "AI" refers to the detector's AUC score on the original AI-generated text.

Dataset	Detector	AI	SICO-Para	SICO-Gen
Writing	GPT2-D	0.848	0.332	0.099
	+ SICO-D	$0.842_{(-0.006)}$	$0.762_{(+0.430)}$	$0.691_{(+0.592)}$
QA	GPT2-D	0.906	0.576	0.489
	+ SICO-D	$0.892_{(-0.014)}$	$0.815_{(+0.239)}$	$0.784_{(+0.295)}$
Review	GPT2-D	0.952	0.264	0.619
	+ SICO-D	$0.949_{(-0.003)}$	$0.782_{(+0.518)}$	$0.834_{(+0.215)}$

6.2 Discussion of AI Detection Arms Race

Considering the evolution of AI detection techniques, the attackers and defenders are actually engaged in an arms race, where both sides continuously develop their own optimal strategies of variability and sophistication to overcome the opponents. We expect the newly proposed attack method will remain effective until the defender collects enough adversarial examples and trains a new detector on them. Then the attacker might develop a new technique to evade the most recent detector. This arms race presupposes that both sides are willing to share information like attack technique or detector API. If either side stops sharing, the race will come to a halt. Practically speaking, the defenders hold a superior position in this race, as they can restrict the access to detector, thereby preventing the attacker from improving their methods. Moreover, the defender, typically a large company, possesses more resources, including financial and computing power. The arms race represents the ultimate question in this field, which is complex and significant. Hence, we only provided a surface-level discussion. A more thorough investigation and experimentation on it are left for future research.

7 Conclusion

In conclusion, we have proposed a novel in-context learning approach, SICO, designed to guide LLMs in generating text that can effectively evade detectors. Our extensive experiments on evasion demonstrate the superior performance of SICO, which significantly reduces the detection capabilities of existing AI text detectors across three tasks. A comprehensive human evaluation shows SICO text can achieve human-level readability and task completion rates.

Looking ahead, SICO could act as a data generator and be integrated during the training phase of AI detectors, which may enhance their robustness. Furthermore, the core concept of SICO, namely, substitution-based in-context learning, could be applied to a variety of text generation tasks. We believe that this opens up new avenues for future research in the fields of text generation and in-context learning.

8 Ethics Statement

The intention of this paper is not to offer a potential method for evading AI-generated text detection systems. Instead, our aim is to raise awareness within the broader community about the vulnerabilities of existing AI-generated text detection systems to such technology. As many LLMs are public available and free to use, many people can adjust their prompt and generate text that evades these detectors. Given the ease of evasion illustrated in this study, these detectors are not robust yet. We hope the research community can stress test their detectors against text generated by carefully crafted prompt, and create more robust detectors in the future.

Besides presenting a potent attack technique, we also offer defensive methods against it. We believe future research will develop more sophisticated methods to enhance the robustness of AI detectors. To support the research in this field, we make our codes and data publicly available.

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